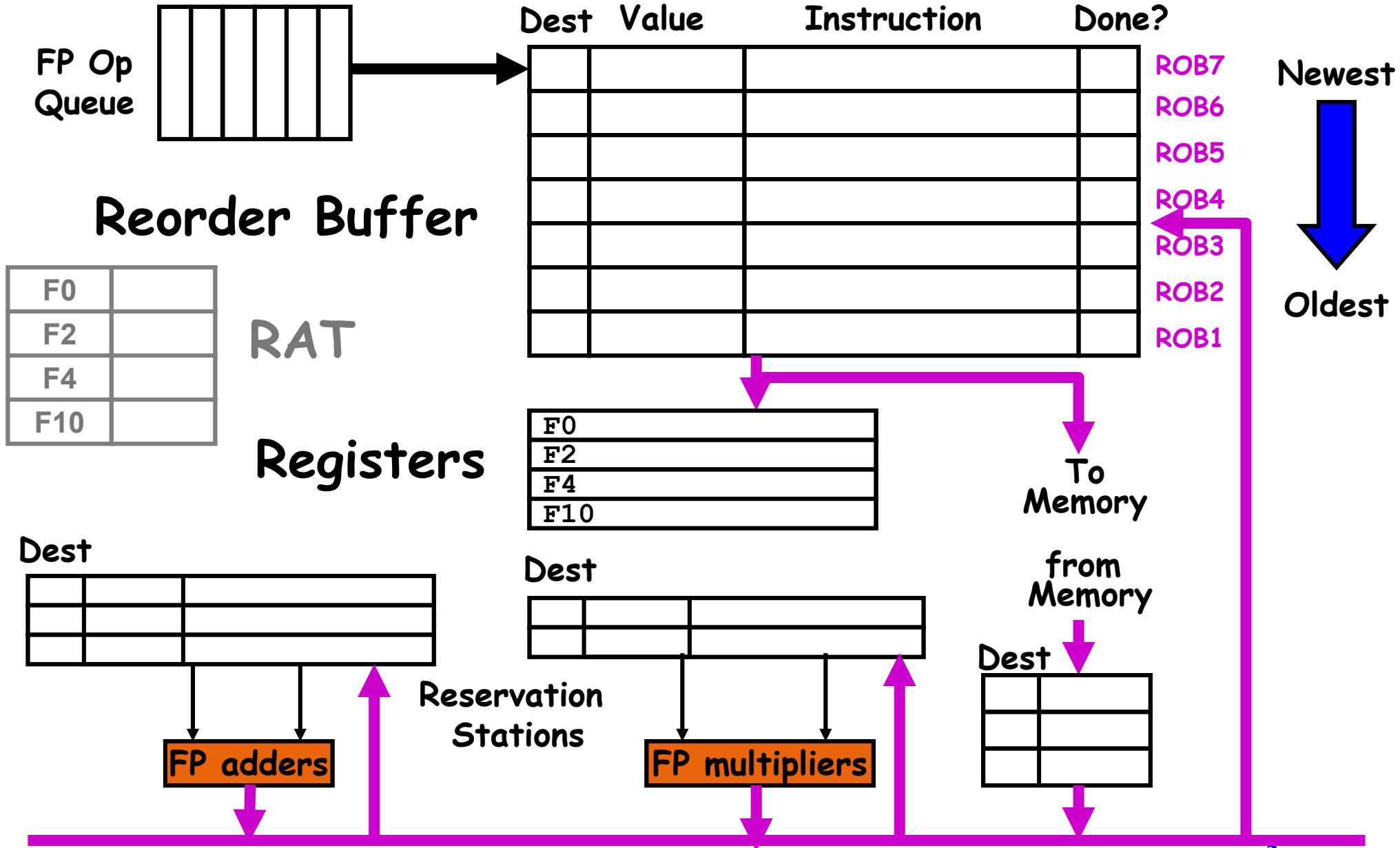


Example of code execution on a processor based on the Tomasulo architecture

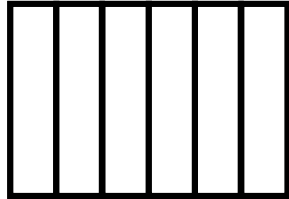
*Slides modified from Prof. Dave Patterson,
Prof. John Kubiawicz, Prof. Nancy Warter-Perez (Berkeley Un.), and
Zvika Guz (Technion)*

Tomasulo With Reorder buffer: architecture



Tomasulo With Branch and Store Instructions

FP Op Queue



Reorder Buff

F0	
F2	
F4	
F10	

RAT

Registers

Dest

FP adders

Reservation Stations

Dest

FP multipliers

ROB1

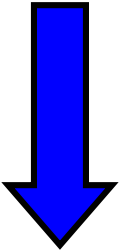
The Register Alias Table (RAT) is responsible for mapping between architectural and physical registers, thus implementing register renaming.

To Memory

from Memory

Dest

Newest

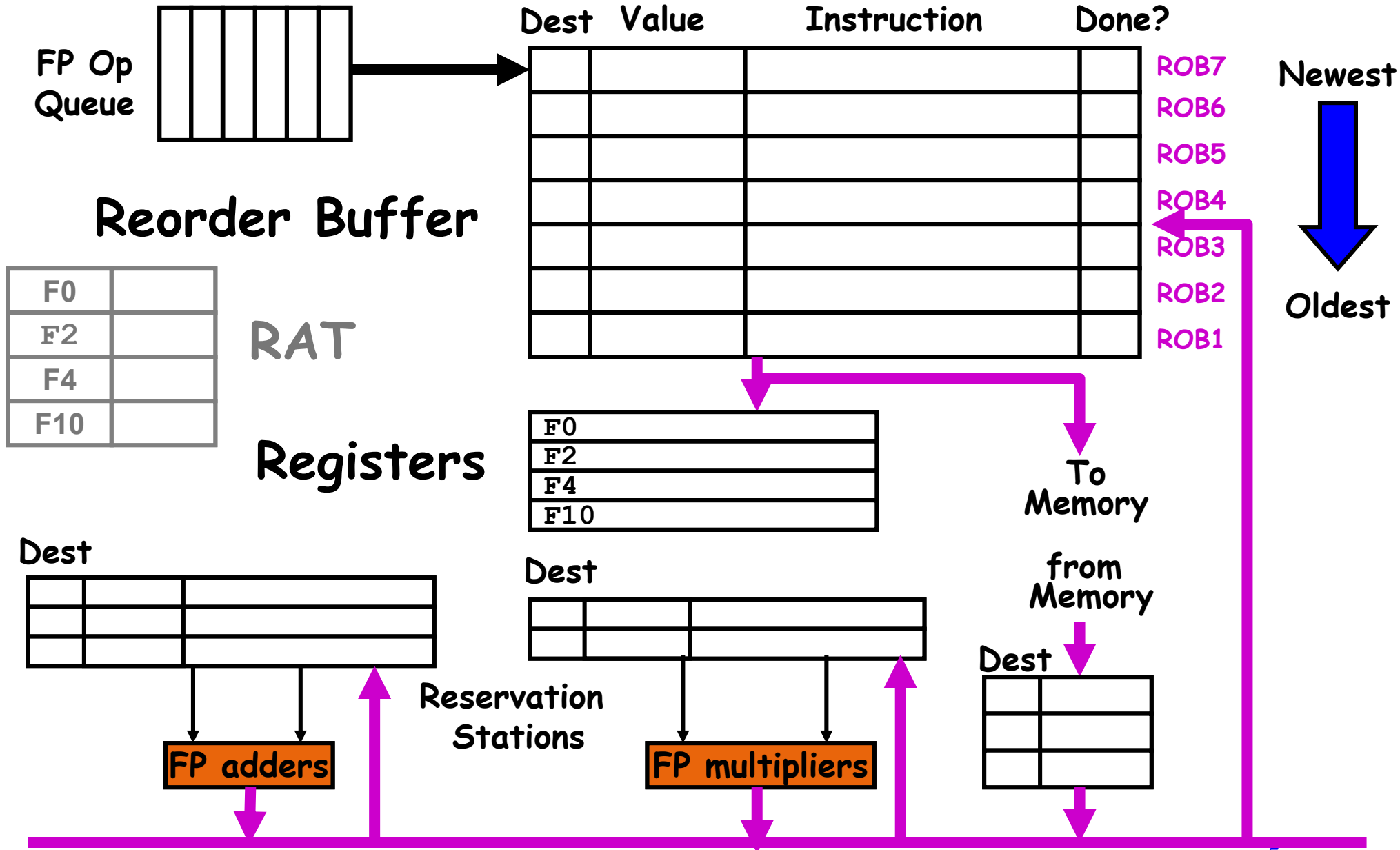


Oldest

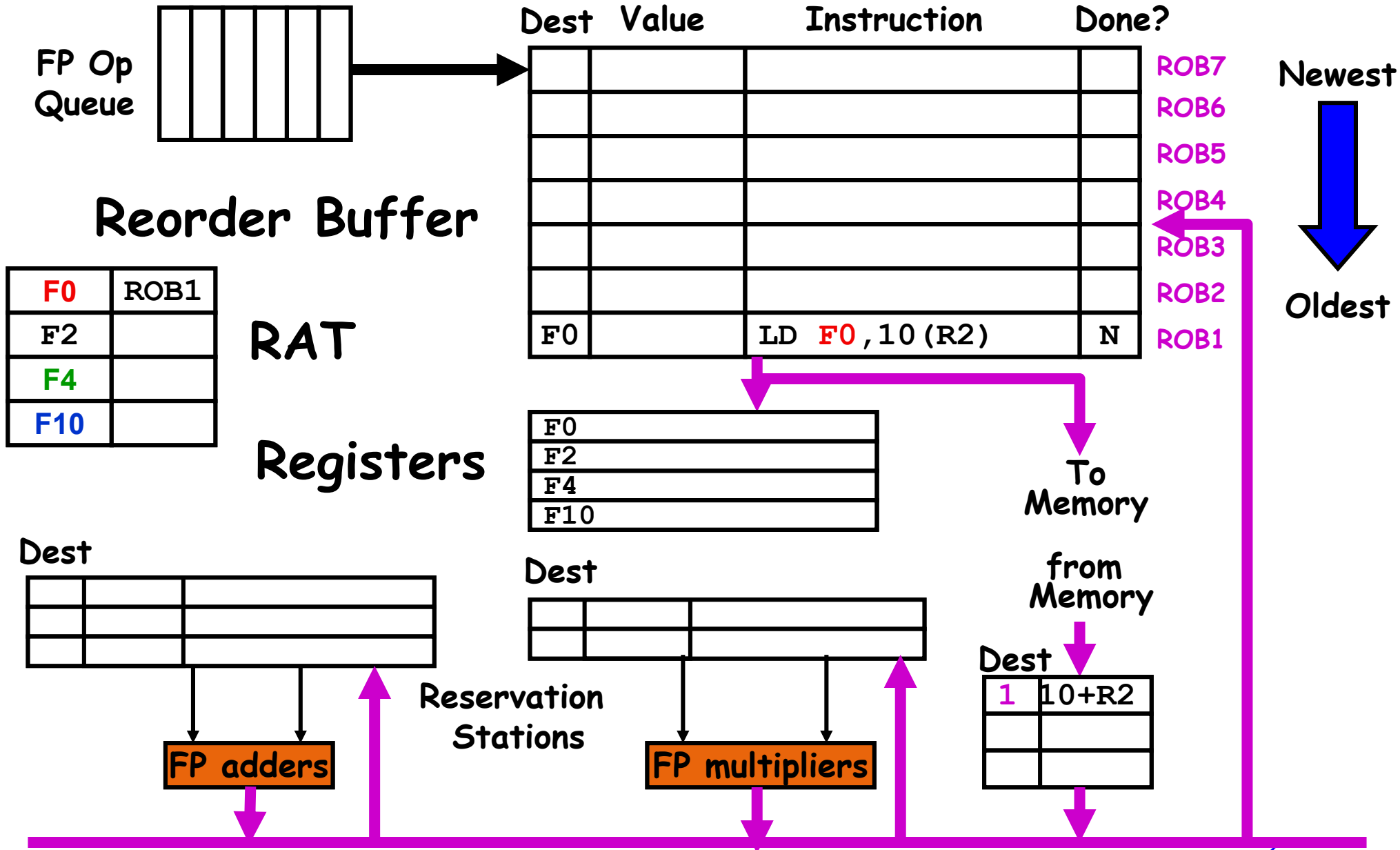
Code example

```
1. LD F0, 10(R2)
2. ADDD F10, F4, F0
3. DIVD F2, F10, F6
4. BNE F2, <...>
5. LD F4, 0(R3)
6. ADDD F0, F4, F6
7. ADDD F0, F4, F6
```

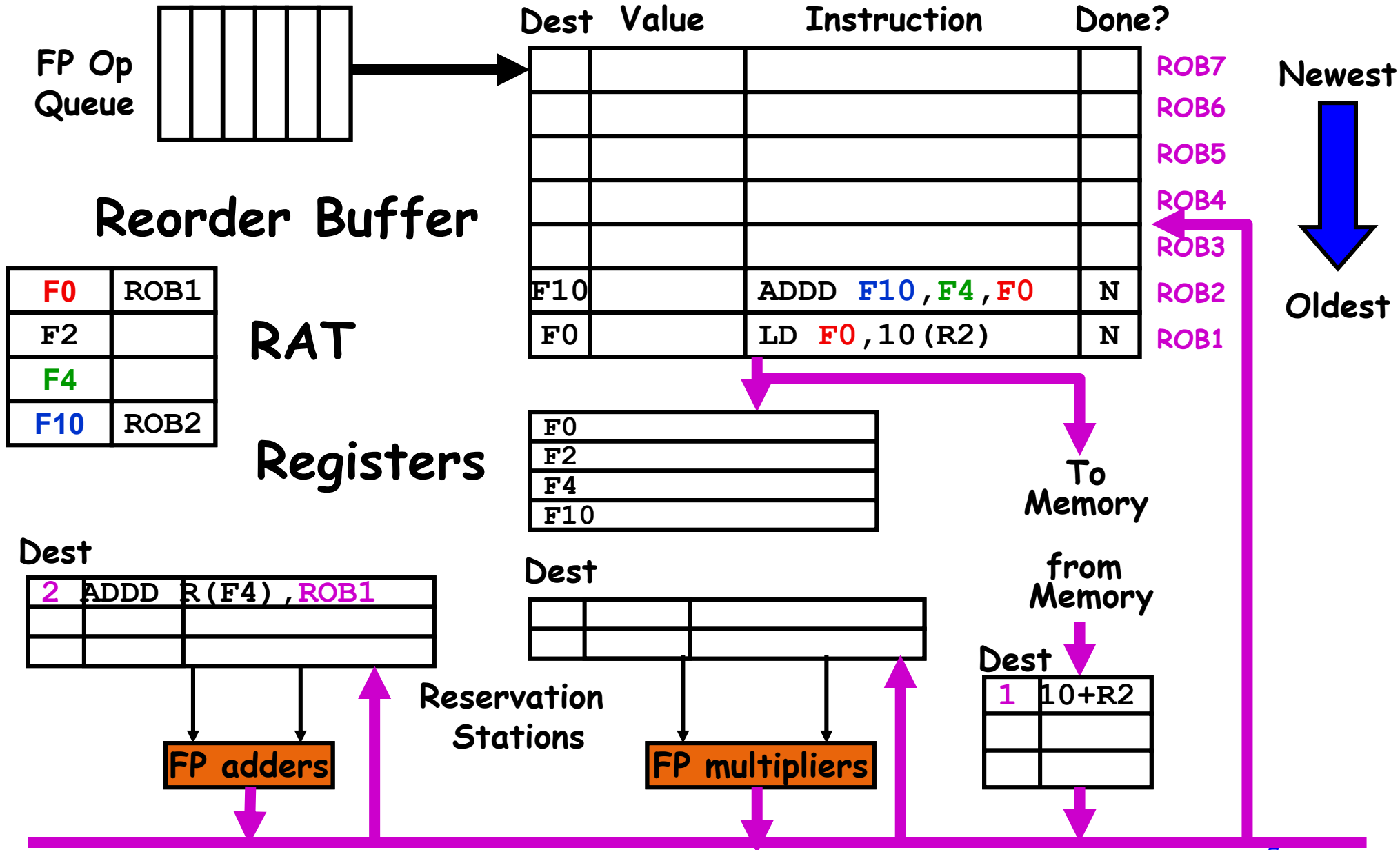
Tomasulo With Reorder buffer



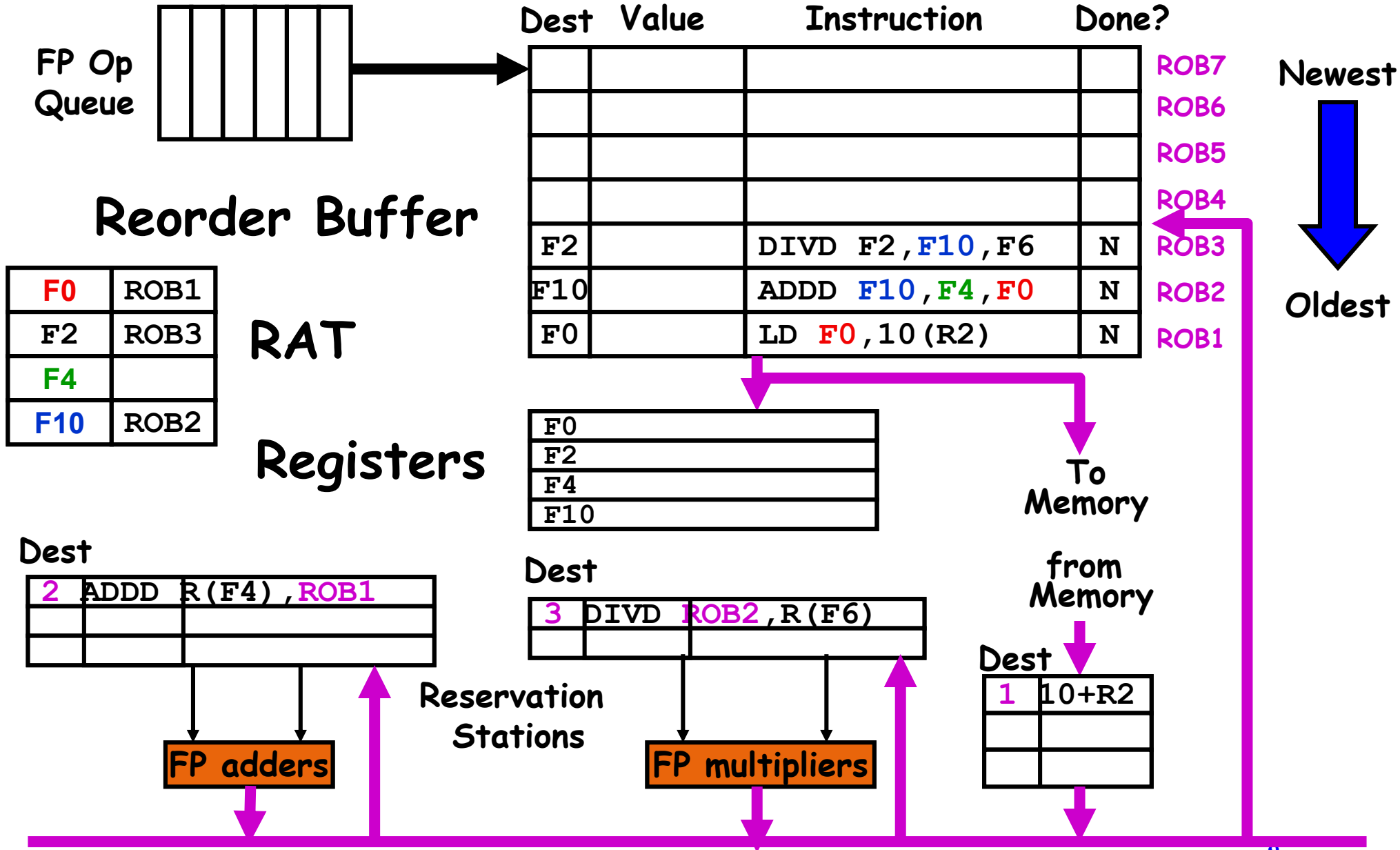
Tomasulo With Reorder buffer: 1



Tomasulo With Reorder buffer: 2

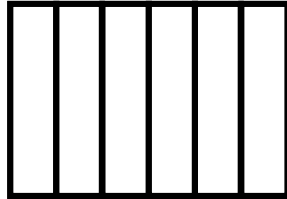


Tomasulo With Reorder buffer: 3



Tomasulo With Reorder buffer: 4, 5, 6

FP Op Queue



Reorder Buffer

F0	ROB6
F2	ROB3
F4	ROB5
F10	ROB2

RAT

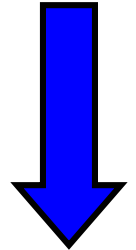
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F4		LD F4, 0 (R3)	N
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

3	DIVD	ROB2, R(F6)

from Memory

Dest

1	10+R2
5	0+R3

Dest

2	ADDD	R(F4), ROB1
6	ADDD	ROB5, R(F6)

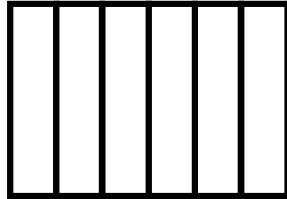
Reservation Stations

FP adders

FP multipliers

Tomasulo With Reorder buffer: 7

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

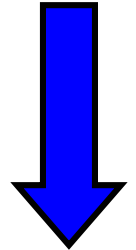
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F0		ADDD F0, F4, F6	N
F4		LD F4, 0 (R3)	N
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

2	ADDD	R(F4), ROB1
6	ADDD	ROB5, R(F6)
7	ADDD	ROB5, R(F6)

FP adders

Dest

3	DIVD	ROB2, R(F6)

Reservation Stations

FP multipliers

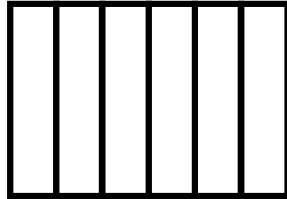
from Memory

Dest

1	10+R2
5	0+R3

Tomasulo With Reorder buffer: 8

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

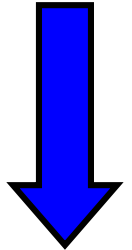
Registers

Dest Value Instruction Done?

F0		ADDD F0, F4, F6	N
F0		ADDD F0, F4, F6	N
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

F0
F2
F4
F10

To Memory

Dest

2	ADDD	R(F4), ROB1
6	ADDD	M[10], R(F6)
7	ADDD	M[10], R(F6)

FP adders

Dest

3	DIVD	ROB2, R(F6)

Reservation Stations

FP multipliers

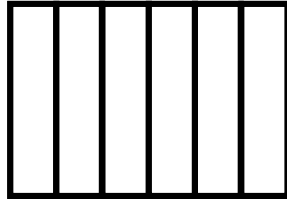
from Memory

Dest

1	10+R2

Tomasulo With Reorder buffer: 9

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

Registers

Dest	Value	Instruction	Done?
F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0		LD F0, 10 (R2)	N

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest
↓
Oldest

F0
F2
F4
F10

To Memory

from Memory

Dest

2	ADDD R(F4), ROB1

FP adders

Dest

3	DIVD ROB2, R(F6)

FP multipliers

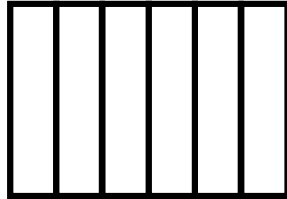
Reservation Stations

Dest

1	10+R2

Tomasulo With Reorder buffer: 10

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

Registers

Dest	Value	Instruction	Done?
F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10		ADDD F10, F4, F0	N
F0	M[2]	LD F0, 10 (R2)	Y

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest
↓
Oldest

F0
F2
F4
F10

To Memory

from Memory

Dest

2	ADDD R(F4), M[2]

Dest

3	DIVD ROB2, R(F6)

Reservation Stations

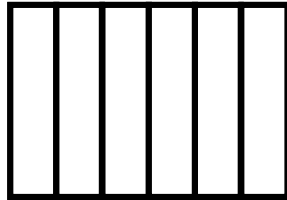
FP adders

FP multipliers

Dest	

Tomasulo With Reorder buffer: 11

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	ROB2

RAT

Registers

Dest

FP adders

Reservation Stations

Dest

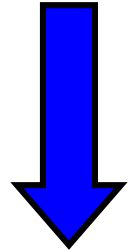
3	DIVD	<val4>, R(F6)

FP multipliers

Dest	Value	Instruction	Done?
F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2		DIVD F2, F10, F6	N
F10	<val4>	ADDD F10, F4, F0	Y
F0	M[2]	LD F0, 10 (R2)	C

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest



Oldest

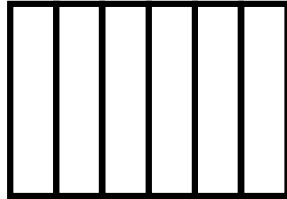
To Memory

from Memory

Dest

Tomasulo With Reorder buffer: 12

FP Op Queue



Reorder Buffer

F0	ROB7
F2	ROB3
F4	ROB5
F10	

RAT

Registers

Dest

FP adders

Reservation Stations

Dest

FP multipliers

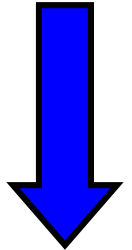
from Memory

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2	<val5>	DIVD F2, F10, F6	Y
F10	<val4>	ADDD F10, F4, F0	C
F0	M[2]	LD F0, 10 (R2)	C

ROB7
ROB6
ROB5
ROB4
ROB3
ROB2
ROB1

Newest

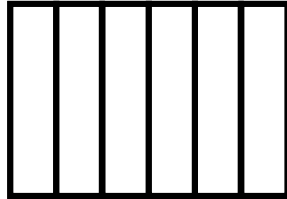


Oldest

To Memory

Tomasulo With Reorder buffer: 13

FP Op Queue



Reorder Buffer

F0	ROB7
F2	
F4	ROB5
F10	

RAT

Registers

Dest

FP adders

Reservation Stations

Dest

FP multipliers

Dest Value Instruction Done?

F0	<val3>	ADDD F0, F4, F6	Y
F0	<val2>	ADDD F0, F4, F6	Y
F4	M[10]	LD F4, 0 (R3)	Y
--		BNE F2, <...>	N
F2	<val5>	DIVD F2, F10, F6	C
F10	<val4>	ADDD F10, F4, F0	C
F0	M[2]	LD F0, 10 (R2)	C

ROB7

ROB6

ROB5

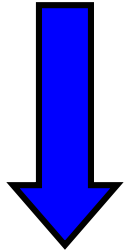
ROB4

ROB3

ROB2

ROB1

Newest



Oldest

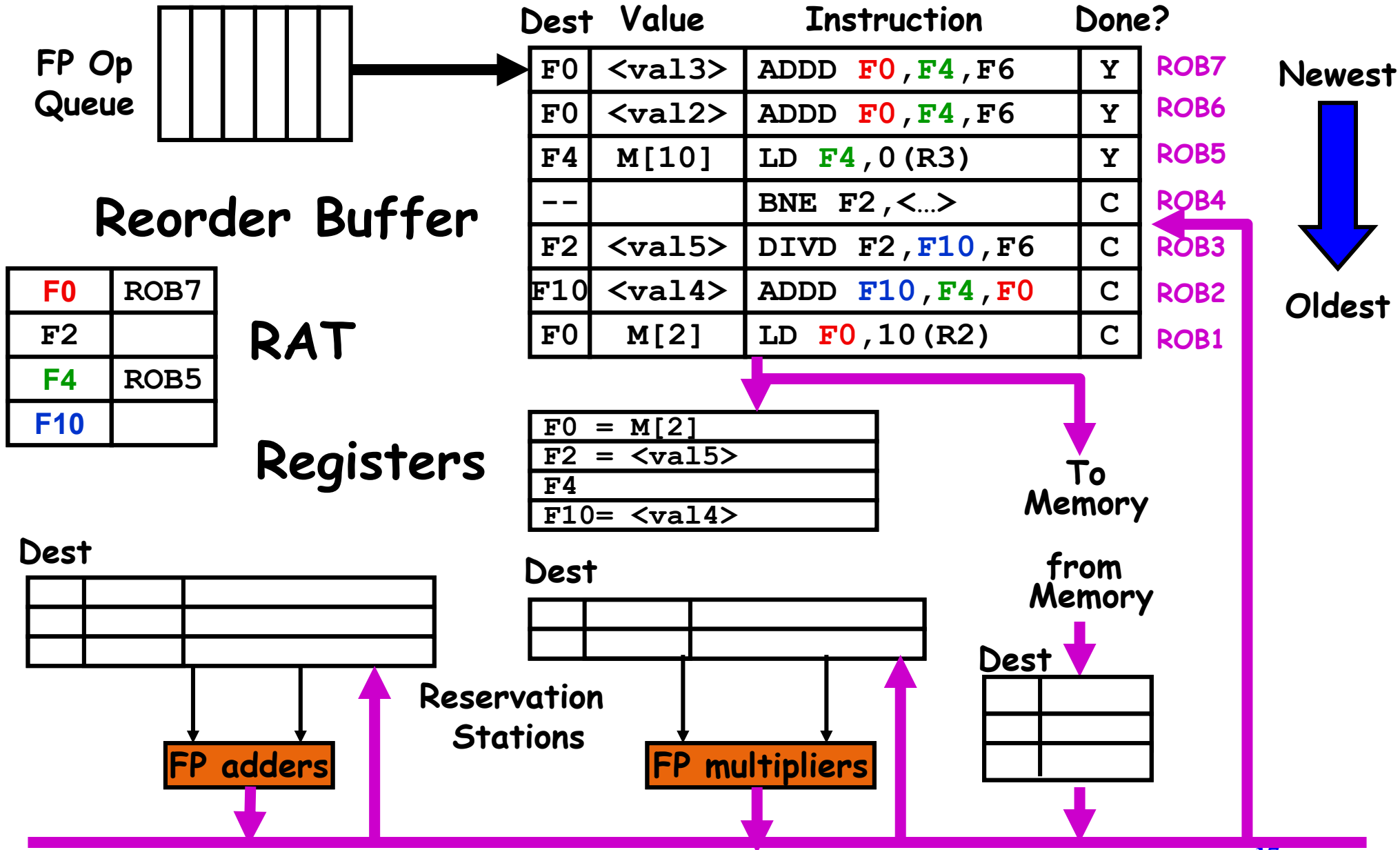
F0 = M[2]
F2 = <val5>
F4
F10 = <val4>

To Memory

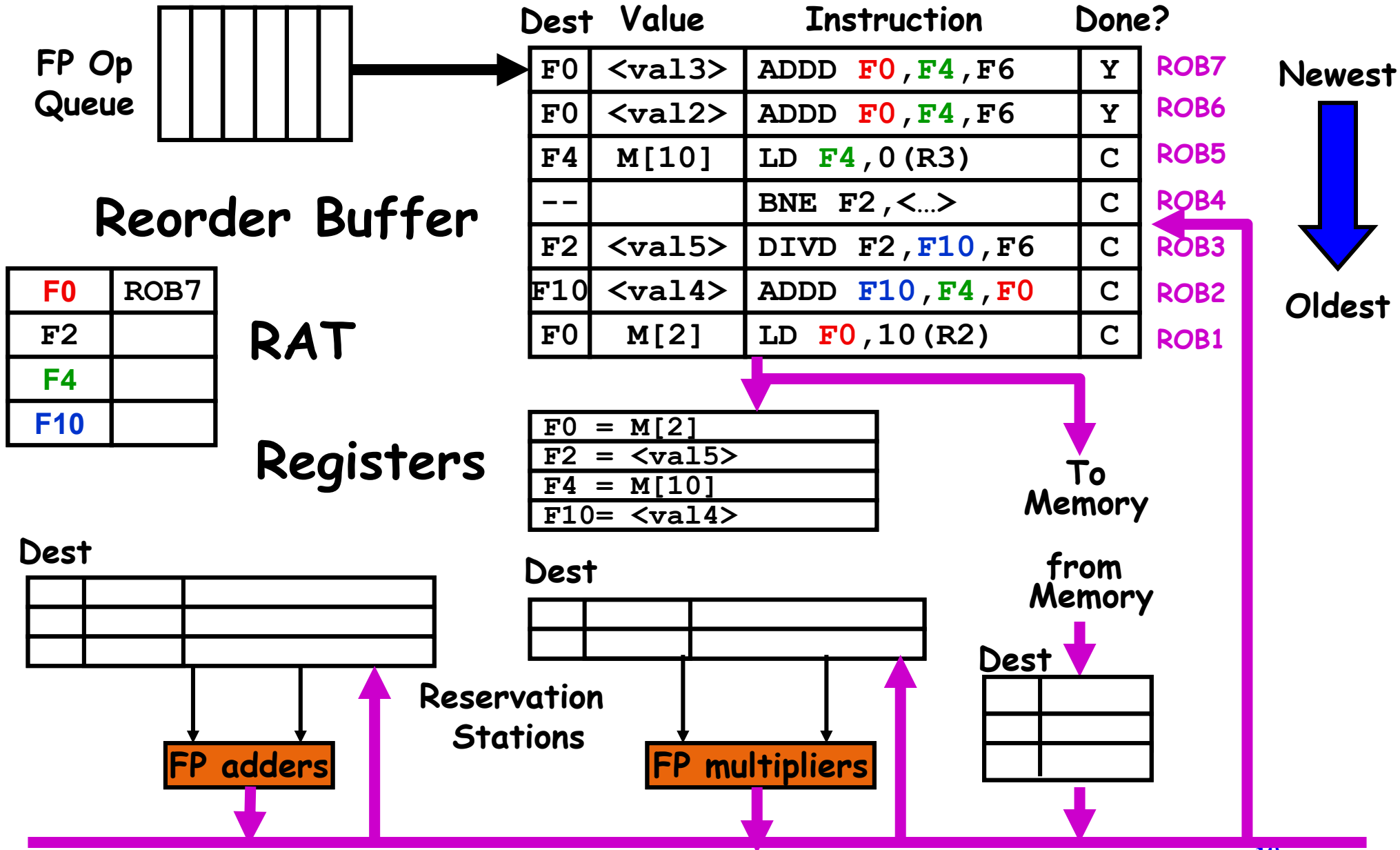
from Memory

Dest

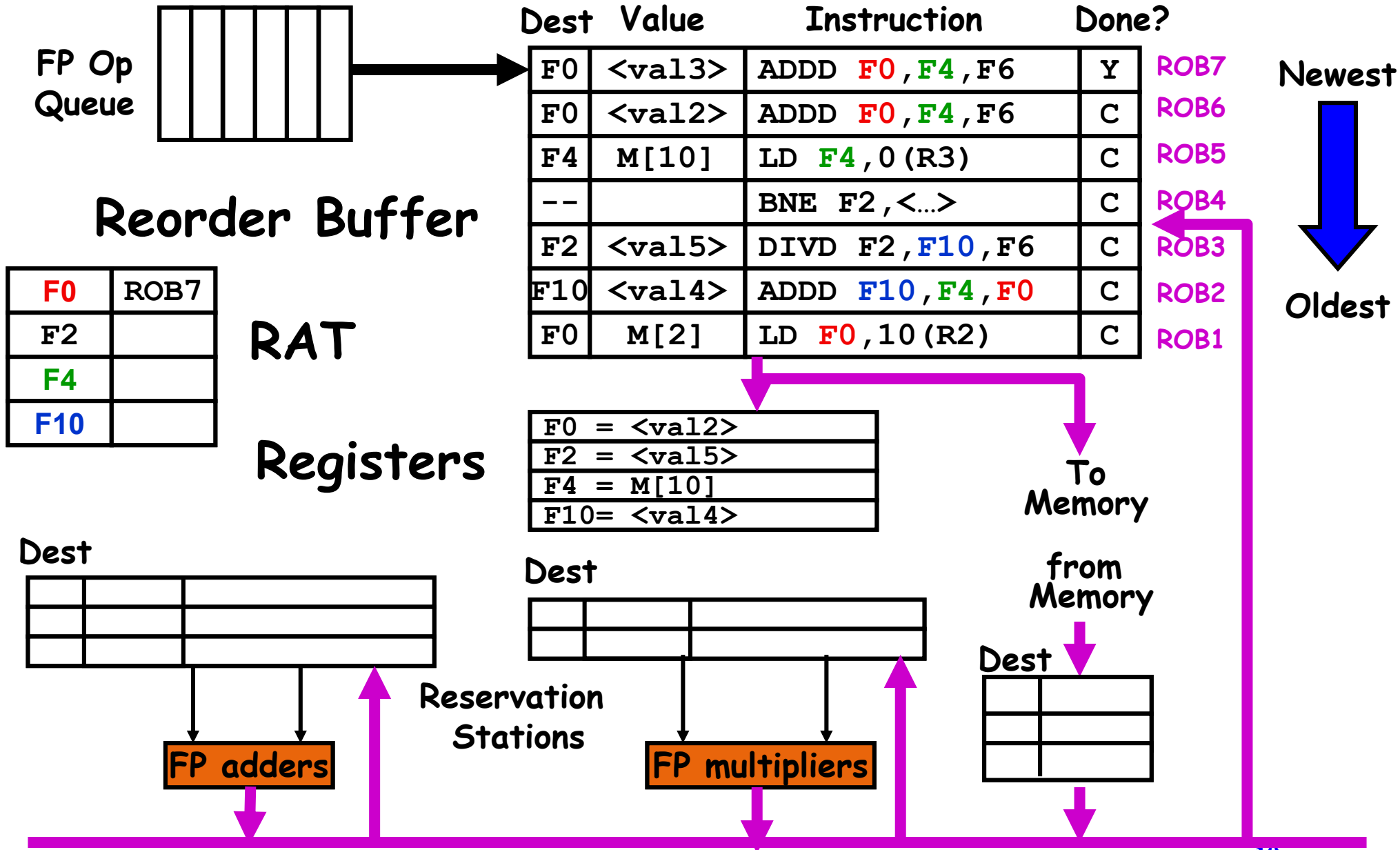
Tomasulo With Reorder buffer: 14



Tomasulo With Reorder buffer: 15



Tomasulo With Reorder buffer: 16



Tomasulo With Reorder buffer: 17

